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FISHERIES

Evolutionary adaptation to global change reduces sustainable fisheries yields

Jan Kozłowski[1]†, Dustin J. Marshall[2]†, Craig R. White[2]*†

Global warming is altering the fisheries that underpin food security, but projections of these impacts generally exclude evolutionary processes. We describe a model that forecasts how fish will adapt to future climates and the consequences of that evolution for fisheries yields. We predict that fish in warmer waters will grow faster but evolve earlier maturation, decreasing their maximum size. We predict that evolution ameliorates the impacts of climate change on fish fitness but exacerbates its impacts on fisheries yields—worsening losses by ~50%. Excluding evolution overestimates future yields under all emissions scenarios, but evolution's impacts are greatest under the most extreme scenarios. All life histories may evolve in response to global change—this evolution should be considered in projections of ecosystems and their services.

Climate change is altering the biological processes that underpin society and biodiversity (*1*, *2*); anticipating these alterations is essential for any effort to mitigate their impacts. Fisheries provide billions of people with animal protein, demand for which is predicted to only increase (*3*). Global warming therefore threatens not just ecosystems but also global food security. To anticipate this threat, a swathe of mechanistic models has sought to predict how fished species will respond to warming in the absence of evolutionary change (*4*–*11*). These mechanistic models tend to assume that fish are evolutionarily inert, modeled to respond in the future as they do now. But these contemporary responses may be poor proxies when fish adapt to very different future conditions. In other words, if fish evolve in response to changing thermal regimes (*12*, *13*), then models should reflect this biological reality (*14*, *15*).

Predicting exactly how fish will adapt to warmer temperatures is a formidable challenge because any number of traits may evolve with temperature in idiosyncratic ways. One unifying factor is mortality, a powerful selective force in life-history evolution (*16*–*25*). Warmer temperatures are known to increase mortality rates (*26*, *27*) and decrease lifespans (*28*). Thus, the strong covariances between temperature, mortality rate, and life history offer a framework for predicting how fishes (and fisheries yields) may change in warmer oceans. This life-history optimization approach departs from mechanistic approaches in that it centers, rather than excludes, the role of adaptation in modifying the impacts of global change on future fisheries [e.g., (*23*)] (fig. S1).

Here we explore how warming may alter the productivity of the world's fisheries, using a new life-history model based on the principle of optimal resource allocation (*16*–*18*, *23*, *24*) (see fig. S2 for our workflow). We first validate the model by testing its capacity to predict (i) known patterns in trait covariation across nearly 3000 species of fishes (*29*) and (ii) the observed changes in life histories with recent climate warming across the globe (*30*–*33*). We then use this validated model to predict how fish life histories could change under future climate scenarios, on the basis of known relationships between temperature and mortality rate. Finally, we estimate how life-history evolution may alter sustainable harvests for 43 of the world's largest fisheries.

Optimal resource allocation and the evolution of life histories

Animals that exhibit indeterminate growth can be assumed to allocate resources to production (P = growth + reproduction) at a rate described by the equation $P = aw^b$, where w is body mass and a and b are the coefficient and exponent of production, respectively. All production is allocated to growth until maturity and thereafter is divided between growth and reproduction. All the while individuals experience mortality, removing them from the population over time.

Maturing at a smaller size confers the benefit of reproducing before much mortality risk has been accumulated, but maturing at larger sizes confers greater reproductive output (*34*, *35*). The timing of maturity and the optimal balance between growth and reproduction depend on the benefits of growing versus the risks of dying. The life history that maximizes the lifetime production of offspring for a given mortality regime will be favored.

We modeled growth and reproduction by dividing time into productive and nonproductive periods (Fig. 1, A and B). During the productive period, immature animals allocate surplus resources to growth, whereas mature animals allocate surplus resources to both growth and reproduction until a time when allocation to growth ceases (*17*, *18*). When resources are devoted to growth, the increase in size is calculated from integration of $P = aw^b$ as

$$w(t) = \left[a(1-b)\Delta t + w(t-1)^{1-b}\right]^{\frac{1}{1-b}} \quad (1)$$

where Δt is the duration of the growing phase within the productive period. Within each productive period, the reproductive phase occurs after the growth phase. Before maturation, Δt is equal to the total duration of the productive period; Δt is equal to zero once growth has stopped (e.g., Fig. 1). Resources allocated to reproduction during each productive period are invested in synthesizing gametes and the gametes themselves (*36*). For the range of mortality regimes experienced by fish in nature, we determined trajectories of growth and reproduction that maximized fitness by optimizing the value of Δt within each period, beginning at an arbitrary initial size of w_0 = 6 mg (*37*) until an age where the probability of survival fell below 0.001 (*38*).

The model makes no assumptions about when reproduction should begin, the trajectory of growth, or the size at which growth will cease—these are determined only by the optimization of Δt (e.g., Fig. 1). From this optimization, a familiar pattern emerges: Animals allocate ever fewer resources to growth and ever more to reproduction as they accumulate mass. This strategy yields a growth trajectory that resembles the von Bertalanffy growth equation (*39*) but is driven by different underlying processes (*17*). We tested the predictive power of the model by generating growth curves from Eq. 1 for nearly 3000 species (e.g., fig. S3) and comparing these with observed fits to data (Fig. 2).

Life-history optimization in response to different mortality rates recapitulates much of the observed diversity in fish life histories. Our life-history optimization model revealed that the (co)variances across a range of traits in fish are defined largely by their mortality rates (*29*) (Fig. 2 and fig. S4). Fish in higher-mortality regimes grow quickly but mature sooner, and at a smaller size, whereas fish in lower-mortality regimes grow slowly but mature later, and at a larger size (Fig. 2). It is important to note that substantial variation exists apart from the general trends that our model predicts—obviously fish life histories do not evolve in response to mortality rates alone. Nevertheless, our model explains between 58 and 97% of variation in critical life-history traits (Fig. 2 and fig. S4). Based on the success of the model in predicting observed patterns, we then used it to predict how fishes are likely to evolve in response to the higher mortality rates associated with recent and future warming.

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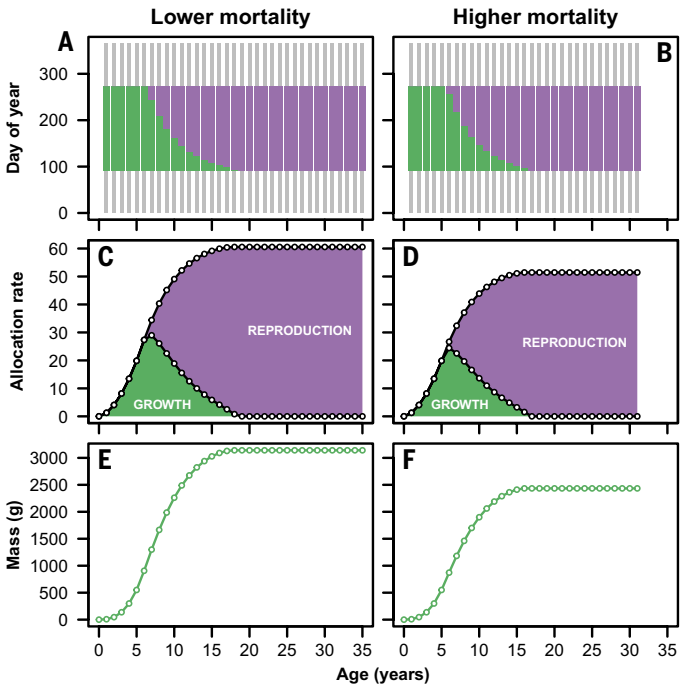

Fig. 1. Mortality alters optimal schedules of allocation to growth and reproduction. (**A** to **F**) Schematic showing optimized growth (green) and reproduction (purple) phases for a hypothetical species ($a = 0.35$, $b = 0.64$ in the production equation $P = aw^b$, where w is body mass; $c = 0.010$, $d = 2.90$ in the weight-length equation $w = cl^d$, where l is body length) under lower mortality [$M = 0.20$; (A), (C), and (E)] and higher mortality [$M = 0.22$; (B), (D), and (F)]. The annual cycle [(A) and (B)] is divided into a nonproductive period (stylized as thin gray bars) and a productive period (thicker colored bars). Within the productive period, growth and reproduction phases are shown as thicker green and purple bars, respectively. Optimization of growth and reproduction for this species under the two mortality regimes alters the switching between growth and reproduction [(A) and (B)], which alters the rate at which energy is expended on growth and reproduction [(C) and (D)] and yields the growth trajectories shown in (E) and (F).

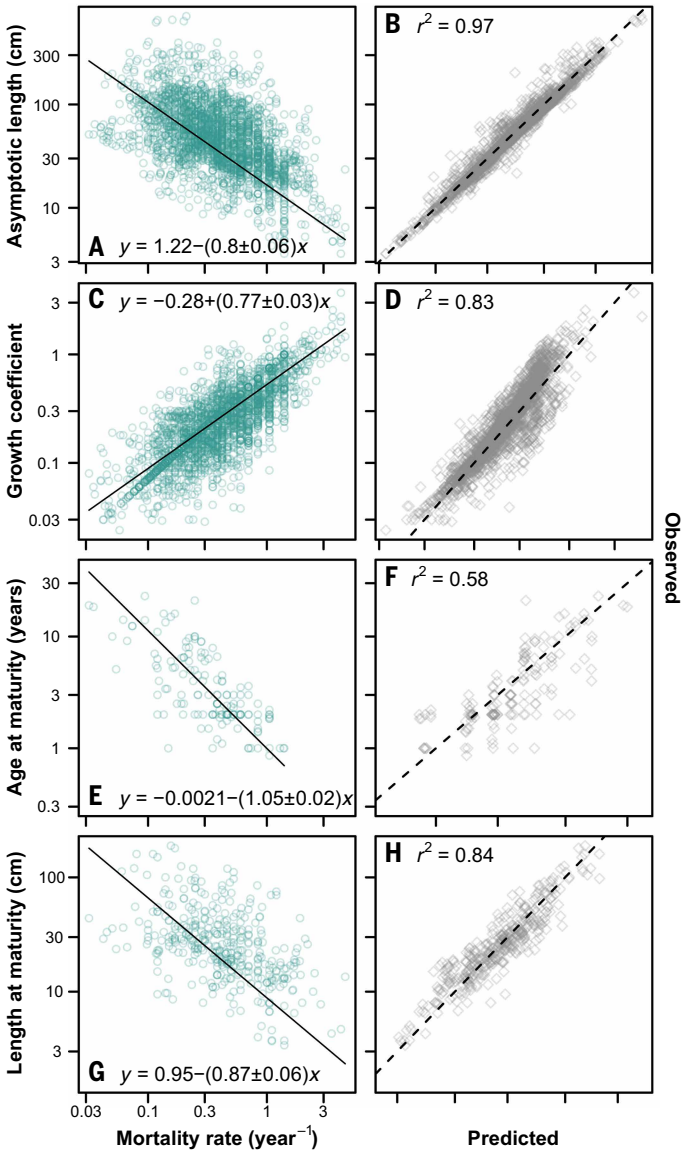

Fig. 2. Evolutionary optimization successfully predicted relationships among mortality and life history. (**A** to **H**) The left column [(A), (C), (E), and (G)] depicts the effect of mortality on (A) asymptotic length (L_∞), (C) the von Bertalanffy growth coefficient (k), (E) age at maturity (t_m), and (G) length at maturity (L_m) for data for fish populations drawn from FishBase; solid lines represent phylogenetically informed relationships fitted to model simulation results with slopes shown ± SE. The right column [(B), (D), (F), and (H)] show the relationships between model predictions and observations from real data for (B) L_m, (D) k, (F) t_m, and (H) L_∞; r^2 is the coefficient of determination for the relationship between observed and predicted; dashed lines are lines of unity.

Life-history optimization in response to recent warming

For most fishes, higher temperatures incur higher natural mortality rates (*24*, *27*). This covariance comes from manifold sources (*12*). We estimated the relationship between mortality rates and temperature for 76 species in nature (*27*): Size-independent natural mortality rates increased by 4.2 ± 1.4% (SE) for every 1°C temperature increase (*38*). Climate warming would be expected to increase mortality rates, which in turn would select for decreases in fish size. On average, our model predicts that warming should decrease maximum length by 3.2 ± 1.1% for each degree of warming (*38*). We tested this prediction against observed changes in fish size associated with recent warming over periods of 11 to ~50 years (table S1). Our predicted size declines match the change in maximum length observed across dozens of species and 1200+ populations (*31*, *32*) (~2 to 5% °C^{-1}), as well as shifts in mean body length (~5% °C^{-1}) observed in 1582 species (*33*). Similarly, we predict that the optimum length at maturity is likely to decrease by 3.5 ± 1.2% °C^{-1}, which emulates observed decreases of ~3% °C^{-1} for 16 species of Mediterranean fish (*30*). Although such changes could be achieved by phenotypic plasticity alone, evolutionary changes depend on the heritability of body size and maturation schedules (*40*). We therefore compiled >300 estimates of genetic variation in body size and the timing of maturation for ~70 species of fish (table S2), as well as studies that observed evolutionary responses to artificial selection on body size (table S3). These compilations are consistent with our model predictions that relatively rapid evolutionary changes in body size are feasible by the end of the century (*38*).

Body size reduction is regarded as a near-universal response to climate change (*41*), yet the ultimate evolutionary explanation for why animals shrink has remained elusive (*24*, *42*, *43*). We find that this shrinking, and the "temperature-size rule" more generally, can be interpreted through the lens of life-history optimization: Fish shrink in warmer (riskier) waters because they accelerate their maturation

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schedules. Differences in the thermal dependence of growth and differentiation (*44*) might then offer a proximate explanation for how warming decreases size. Nevertheless, key uncertainties remain. For example, species will both shift their ranges (*45*) and evolve in response to warming: The relative contribution of both these processes, and how they will alter life histories in combination, remains unclear.

Life-history optimization in response to future warming

The life-history optimization model successfully captures both current day trait (co)variation in fish life histories and the changes to those life histories mediated by recent warming. On the basis of these validations, we then used the model to predict how fish life histories may change in the future. Our model predicts that fish will be ~20% smaller on average in warmer climates, but the magnitude of this change depends on fish habit and habitat (fig. S5). Because freshwater systems are likely to warm the most, size reductions are predicted to be most severe in these habitats (fig. S5). By contrast, there is generally less warming predicted in the Antarctic, so fish are predicted to shrink less there. These projections probably underestimate fish body-size declines because we exclude the impacts of mass mortality events, which are increasing in frequency (*46*). Any additional increase in mortality, whether from disease, deoxygenation, or harvesting, will select for even earlier maturation and smaller sizes. Nevertheless, our conservative estimates of size reductions on the basis of warming alone are predicted to have pronounced impacts on future fisheries yields.

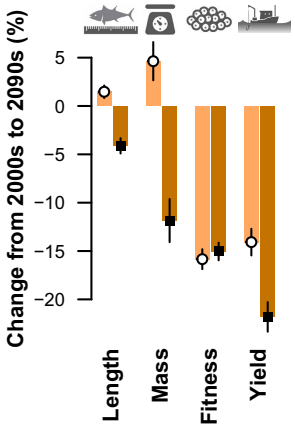

Fig. 3. Adaptation helps fish while harming fisheries. In the absence of an evolutionary response, rising temperature and mortality were predicted to increase the maximum size (length and mass) of fish, but decrease fitness and fisheries yield (lighter bars, unfilled circles). Adaptation reduced the fitness penalty, but reduced body size, and fisheries yields (darker bars, filled squares). Error bars represent approximate 95% confidence intervals, calculated as 1.96 × SEM. Adaptation increased fitness for all species (range: +0.1 to +3.9%) and decreased yield for all species (range: −3.9 to −12.3%).

Evolution helps fish while harming fisheries

To estimate how adaptation to global warming affects fisheries yields, we calculated fisheries reference points for 43 species that account for ~32% of wild-caught landings globally (*47*). We compared reference points with and without adaptation under a range of emission scenarios. Under the "middle of the road" (SSP2-4.5) emission scenario, warming by ~2°C increases growth and mortality rates but decreases longevity. In the absence of evolutionary change, these effects combine to reduce fish reproduction by 16% and fisheries yields by 14% (Fig. 3). These projections of lost yield under climate change are comparable to those of other recent studies (*7*, *8*), and like these studies, our model predicts worsening losses with more extreme warming scenarios (Fig. 4).

We then explored the more realistic scenario whereby fish are allowed to adapt and optimize their life history in response to warming. Our model predicts that evolution ameliorates the impacts of warming on fish fitness slightly (compare −15 versus −16%, Fig. 3) but exacerbates reductions in fisheries yields (compare −22% versus −14%, Fig. 3). Species that experience the greatest change in temperature (and therefore mortality) experience the greatest selective pressure to adapt their life histories the most, so their yields experienced the greatest evolutionary penalty (Fig. 4A). For example, walleye pollock (*Gadus chalcogrammus*), representing 3.6% of global catch in 2022 (*47*), is predicted to experience warming of ~+2.7°C and evolve to be 12% smaller at maximum weight. This evolutionary response would reduce pollock fishery yields by ~17%, a loss of approximately half a million tonnes (Fig. 5). This paradox, whereby evolution ameliorates fitness

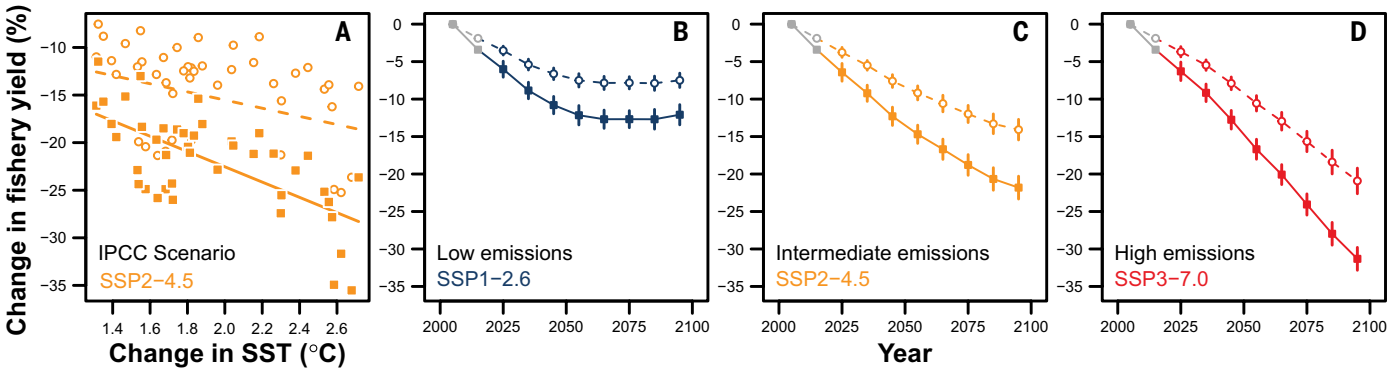

Fig. 4. Species that experience the most warming suffer the greatest reductions in fishery yield. (**A**) For 43 of the world's most fished species, the extent of the reduction in fishery yield depends on the magnitude of warming regardless of whether life histories are optimized (filled squares and solid line, $t_{41} = -5.77$, $P < 0.001$; yield = −6.46 − 8.03 SST; $\lambda = 0.176$) or not optimized (unfilled circles and dashed line, $t_{41} = -3.00$, $P = 0.004$; yield = −6.75 − 4.28 SST; $\lambda = 0.629$). More extreme IPCC scenarios exacerbate predicted losses. Points in (**B**) to (**D**) are means ± 1.96 SE of 43 of the world's most fished species, with changes in yield estimated at a temporal resolution of 10 decadal steps from 2000 to 2100 for three IPCC Shared Socioeconomic Pathways (SSP) representing low emissions [(B) SSP1-2.6: net zero CO_2 emissions occurring in the second half of the century], intermediate emissions [(C) SSP2-4.5: CO_2 emissions remain near current levels until the middle of the century], or high emissions [(D) SSP3-3.7: CO_2 emissions double from current levels by 2100]. The expected warming at the end of the century is ~1.8°C for SSP1-2.6, ~2.7°C for SSP2-4.5, and ~3.6°C for SSP3-7.0. Points in gray are baselines for the 2000s and 2010s.

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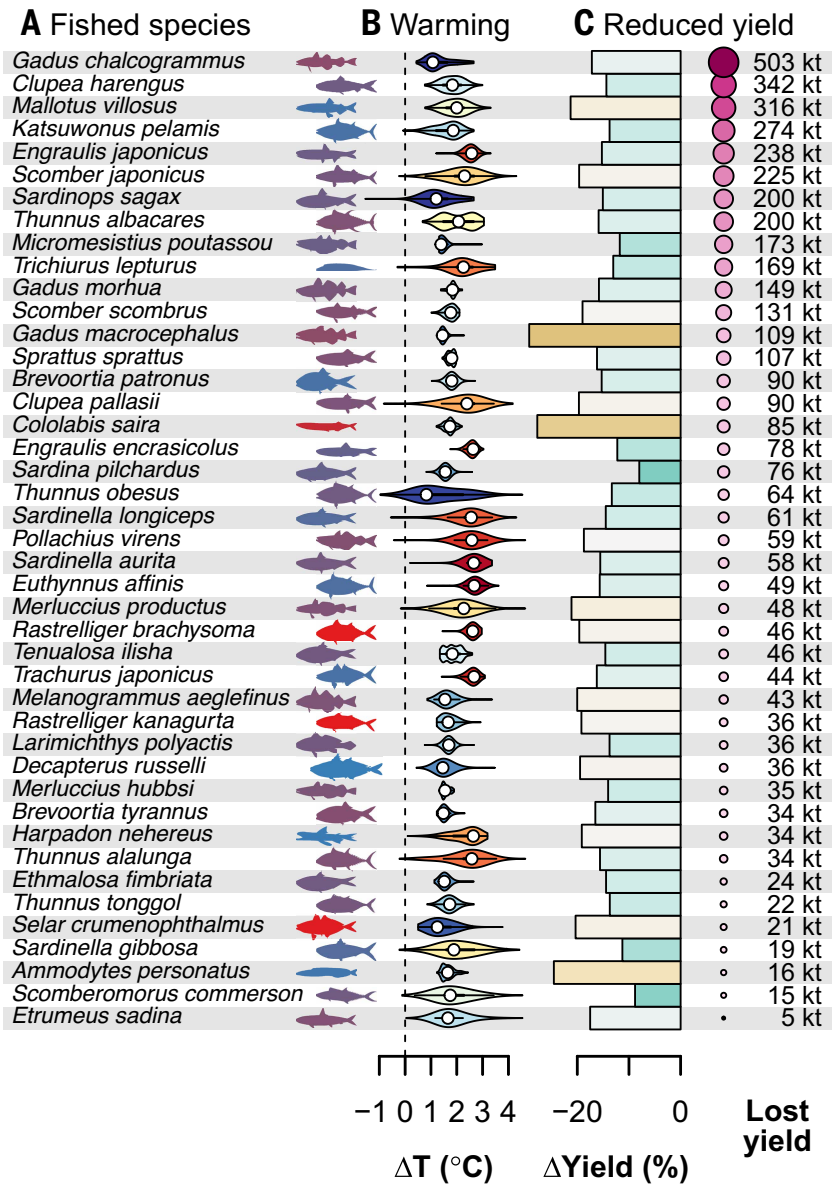

Fig. 5. The effect of climate warming on fish size and fishery yield. (**A**) Forty-three fished species are arranged in order of the magnitude of lost fishery yield; images are colored to represent size changes, with warmer colors representing larger decreases in length. (**B**) Violin plots show the distribution of predicted temperature change estimated over the range of each species from occurrence records (white circles show the median, thick lines the interquartile range, and thin lines extend up to 1.5 times the interquartile range; warmer colors represent larger increases in temperature). (**C**) Changes in fishery yield are represented as change in yield (the difference between yield in the 2090s and in the 2000s) and as lost yield (the product of the change in fishery yield and 2022 FAO landings for each species). Points representing lost yield are scaled such that lost yield is proportional to the area of each circle.

Our model predicts that more extreme climate change scenarios induce greater evolutionary responses in fishes, further reducing yields. Under a high-emissions scenario (SSP3-7.0), in our model fish sizes decreased by ~16%, reducing end-of-century yields by ~30% (Fig. 4D). Thus, good climate policy (SSP1-2.6) could preserve ~18 Mt of fisheries yield each year by forestalling more extreme evolutionary change (Fig. 4B)—another example of the benefits of avoiding worst-case scenarios for the ecosystem services upon which we all rely (*1*, *2*).

Evolutionary impacts of global change beyond fishes and fisheries

Many projections of the impacts of global warming across different taxa and systems exclude evolutionary change. Such models would fail to predict any of the observed evolutionary changes in body size caused by temperature-independent shifts in mortality rates [e.g., (*19*, *20*, *22*, *25*)], whereas our model anticipates such changes (e.g., Fig. 1). We focus on fishes because they have some of the best-resolved relationships between temperature and mortality, but our findings should apply to most organisms. When evolution is considered in the context of global change, it is usually regarded as a process that will increase the likelihood of species persistence. That species will adapt to climate change is probably positive for persistence, but its consequences for ecosystem services may not be.

We suggest that projections of the impacts of global change on species distributions and abundance include a more explicit role of life-history evolution. It is well recognized that warming accelerates ectotherm life histories directly through its effects on physiology. Our framework predicts that—prior to evolutionary optimization—warmed species may increase in size if they grow faster but mature no sooner (Fig. 3). The balance between plasticity and evolution offers a potential explanation for the diversity of temperature-size responses observed in nature (*48*). Our framework also provides a new, indirect path through which global warming could accelerate the pace of life histories: Evolutionary optimization may amplify the physiological effects of warming.

Our focus has been on fisheries, but our model has implications for ecosystems writ large—for example, predators may be affected by changes in the size distribution of their prey. Meanwhile, because life-history optimization alters maturation schedules heterogeneously across species (Fig. 5), such optimization could generate phenological mismatches, disrupting nutrient cycles and trophic cascades (*49*).

penalties but exacerbates fisheries losses, emerged because fish offset the impacts of elevated mortality on fitness by maturing earlier and allocating a lower fraction of resources to growth after maturity. In other words, we expect fish to quicken the pace of their life history as an evolutionary response to elevated mortality [e.g., (*19*, *20*, *22*, *25*)]. Earlier maturation results in smaller body sizes, even though growth rates are higher in warmer waters. Sensitivity analyses suggest that there is no realistic growth scenario that could mitigate the evolutionary impact of increased mortality (figs. S6 and S7). Thus, excluding likely evolutionary responses to global change systematically overestimates future fisheries yields.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S12; Tables S1 to S8; References (*51–215*); MDAR Reproducibility Checklist

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Editor's summary

Anthropogenic climate change is altering environments. Species do respond to such changes, but how quickly this can happen remains unknown. Fish have been shown to evolve rapidly in response to changing conditions, but outcomes are difficult to predict. Koz#owski *et al*. modeled the selective adaptation to warming waters and predicted that fish in these waters will grow faster but will also mature to smaller sizes (see the Perspective by Travis and Reznick). Although the authors concluded that this will help fish survive warming conditions, it will also decrease fisheries yields, with the greatest decreases under the most extreme conditions. Failing to integrate adaptation in fisheries models greatly overestimates yields, leading to reduced sustainability. —Sacha Vignieri

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